LECTURE 7

LAST TIME

LECTURE 6: SUMMARY

Options!

- Options price decompositions
- Put-call parity

THIS TIME

LECTURE 7 OUTLINE

Instruments as portfolios of options

Corporate securities

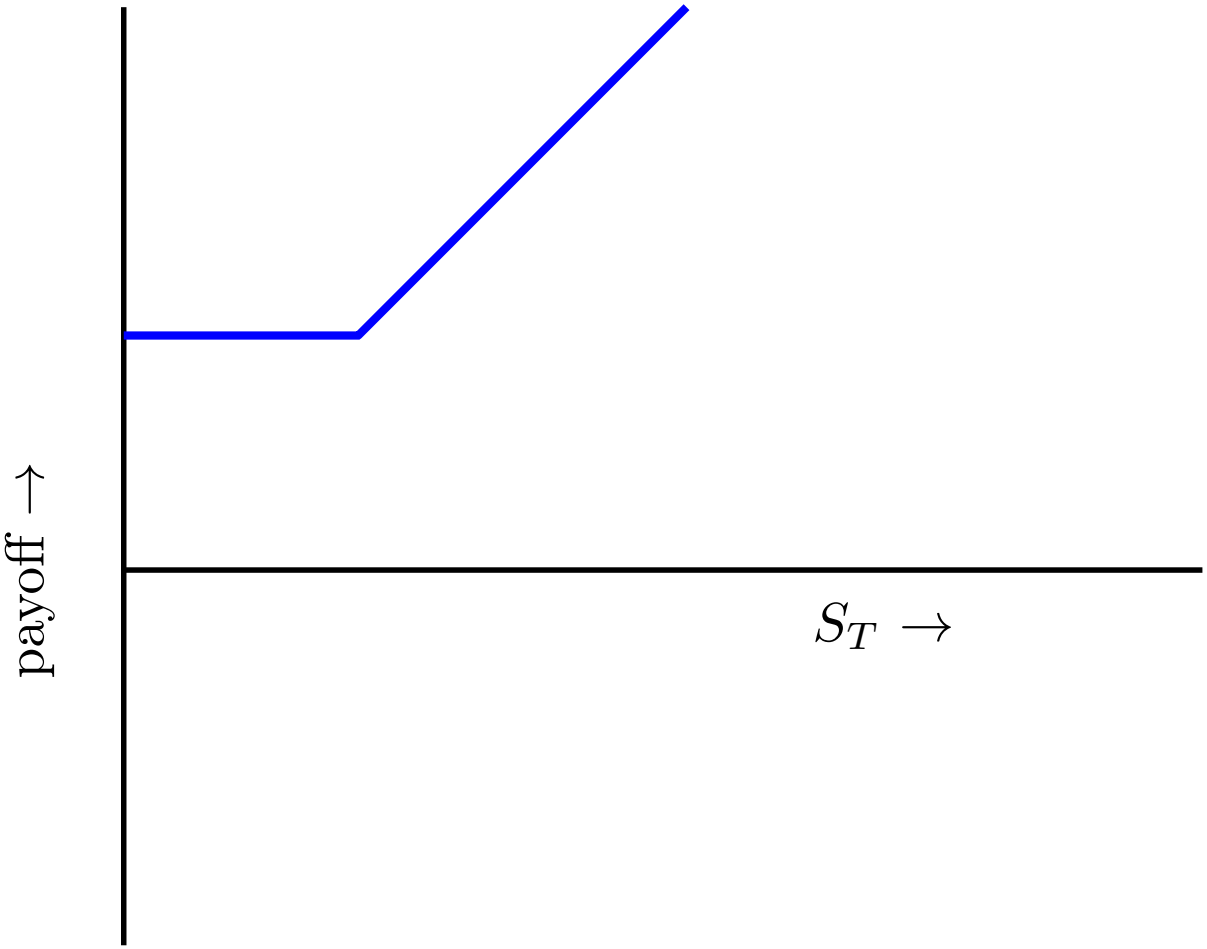
OTHER INSTRUMENTS

PROTECTIVE PUT

You are long a stock but want to protect against downside risk

How can you hedge?

PAYOFF WITH PROTECTIVE PUT

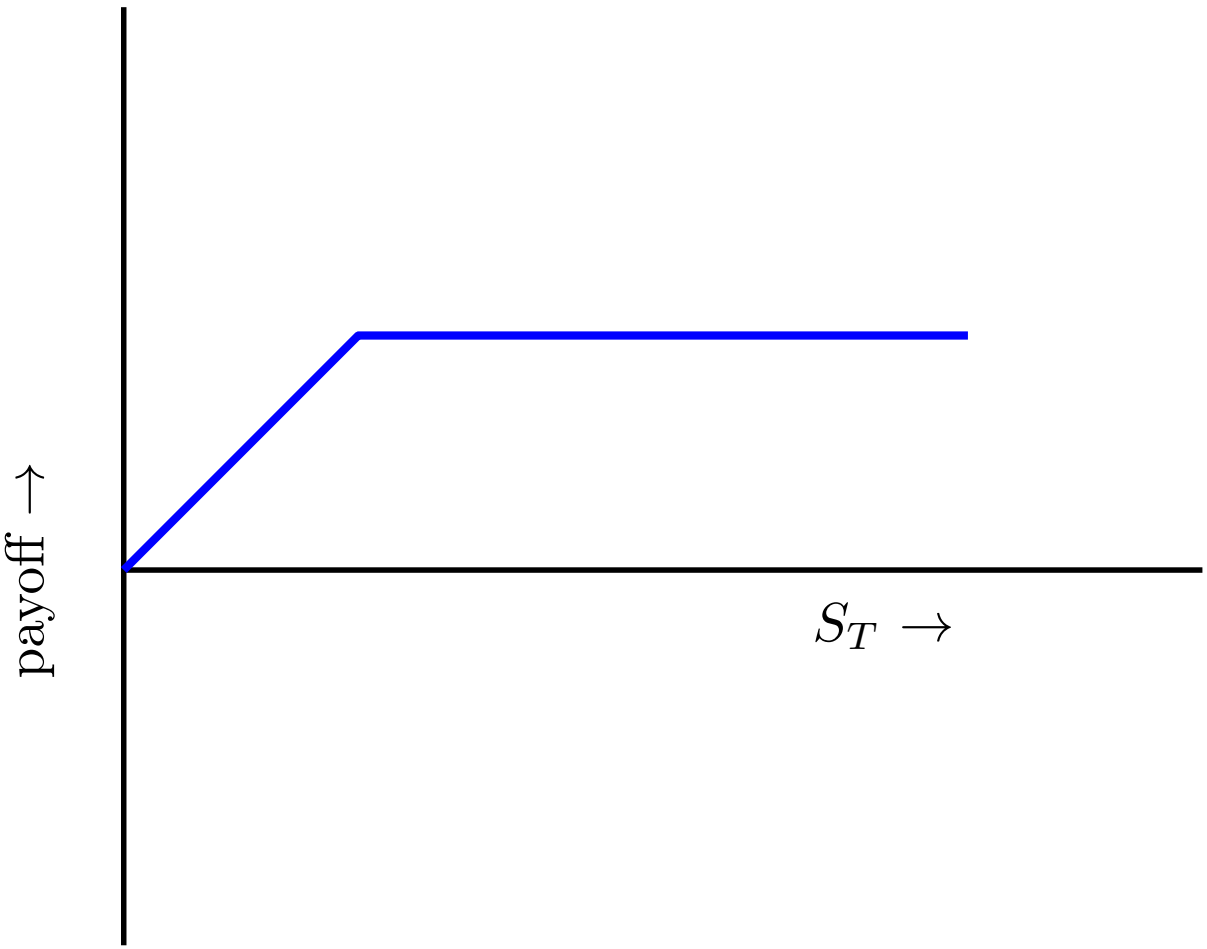


COVERED CALL

You have written a call but are worried about the infinite liability

How can you hedge?

PAYOFF WITH COVERED CALL



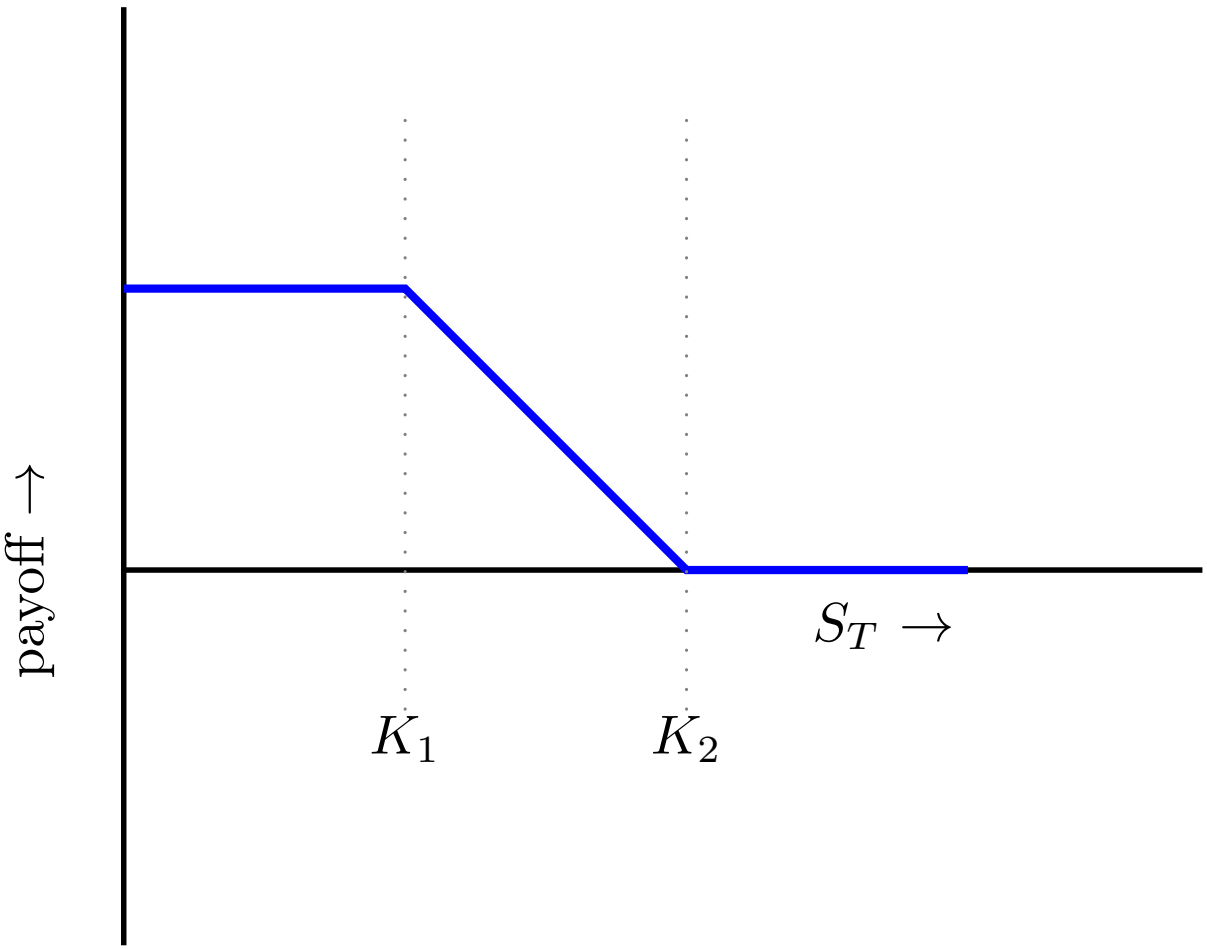
BEAR SPREAD

You are mildly bearish on a stock

How can you speculate on this view?

Hint: do it more cheaply than just shorting

PAYOFF OF BEAR SPREAD



CONSTRUCTING A BEAR SPREAD

The payoff of the bear spread is

$$X_T = \min\left\{K_2 - K_1, \max\{K_2 - S_T, 0\}\right\}$$

You can replicate this with puts

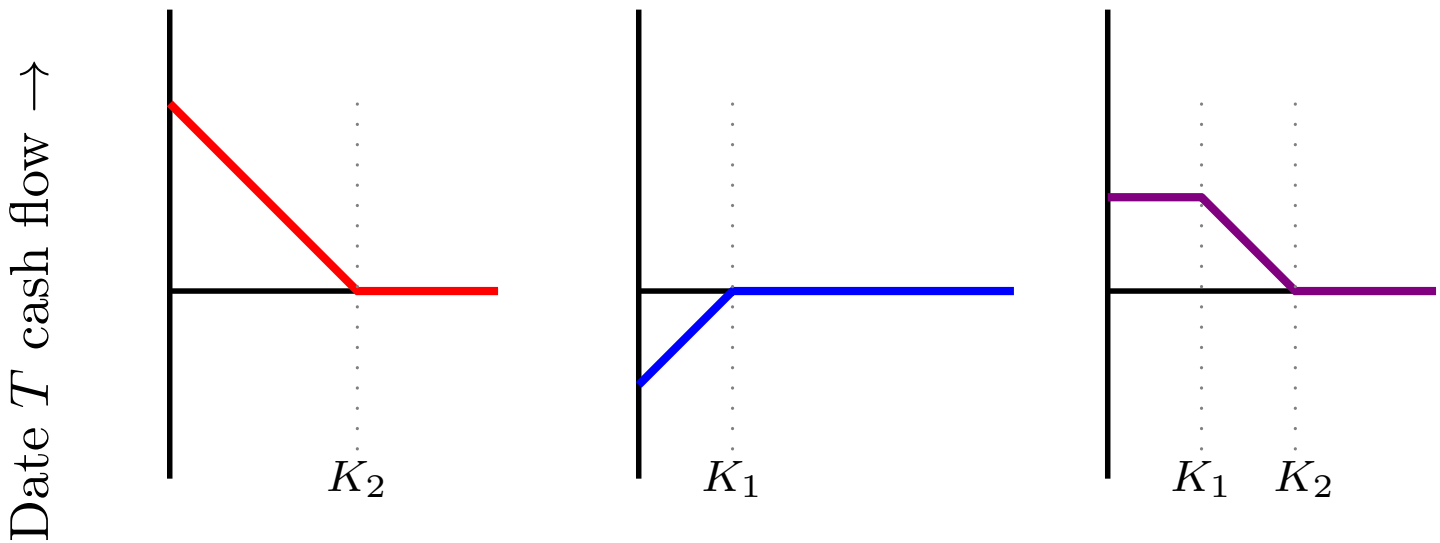
$$X_T = \left(K_2 - S_T\right)^+ - \left(K_1 - S_T\right)^+$$

so

$$X_t = p(t, S_t; T, K_2) - p(t, S_t; T, K_1)$$

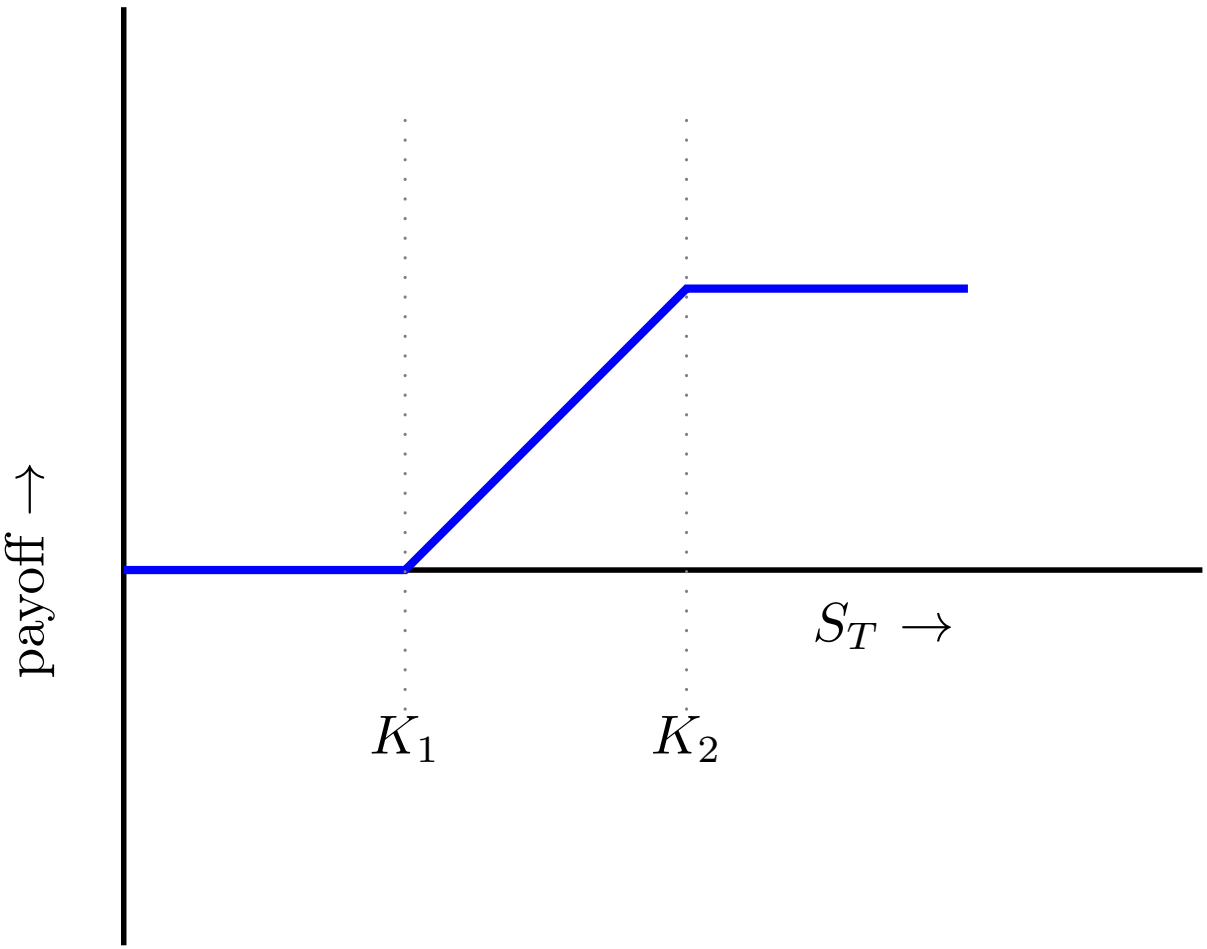
CONSTRUCTING A BEAR SPREAD

long put at K_2 + short put at K_1 = portfolio payoff

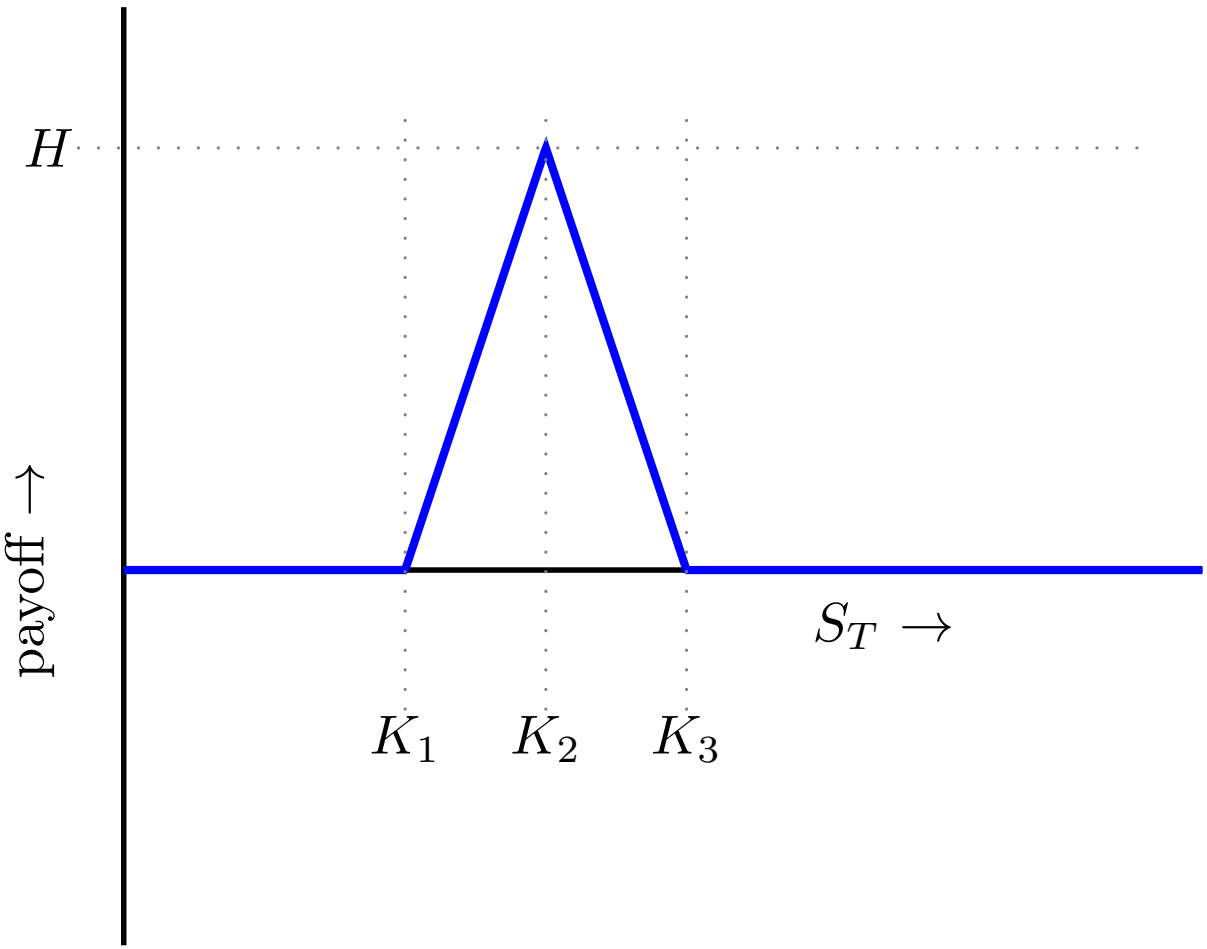


$$(K_2 - S_T)^+ - (K_1 - S_T)^+ = \min\left\{K_2 - K_1, \max\{K_2 - S_T, 0\}\right\}$$

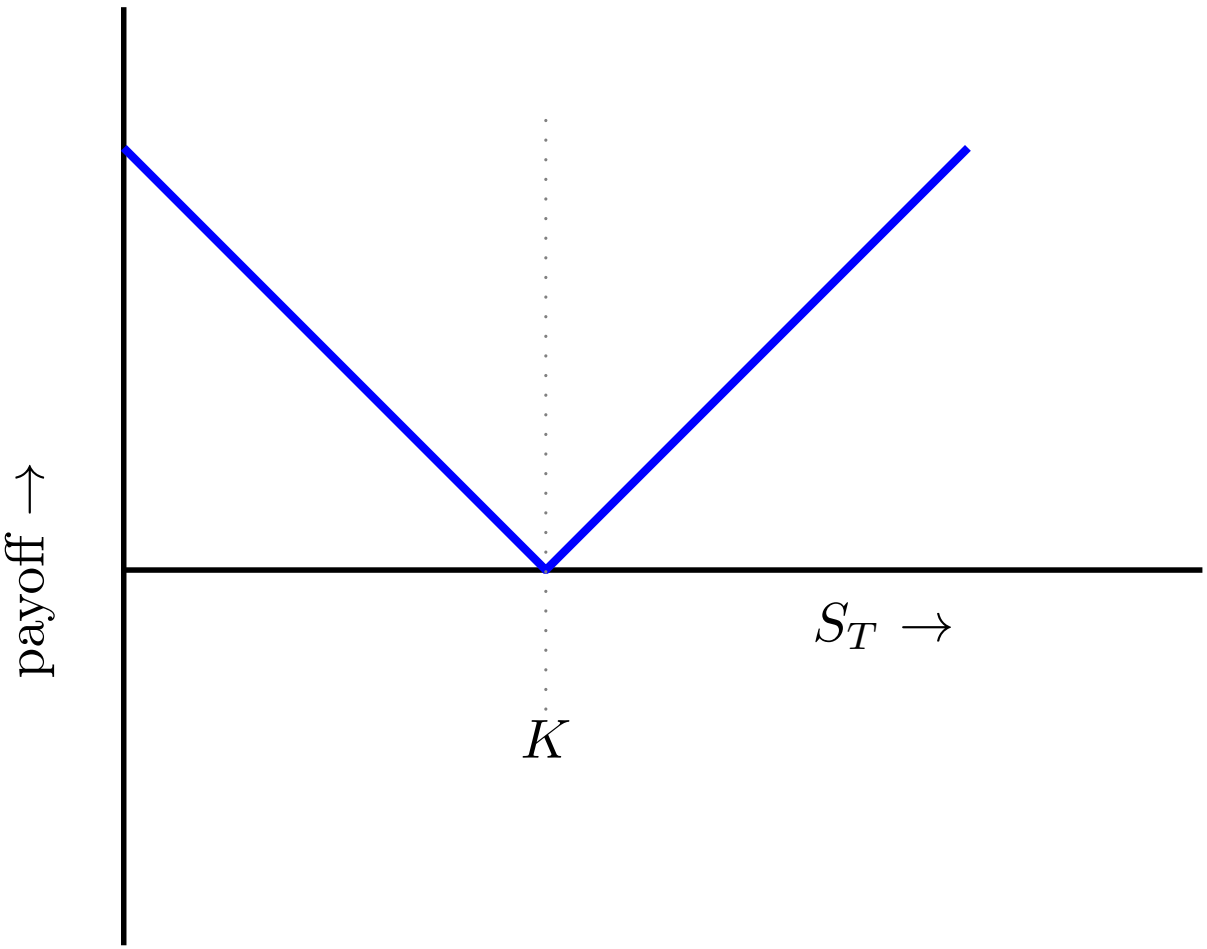
PAYOFF OF BULL SPREAD



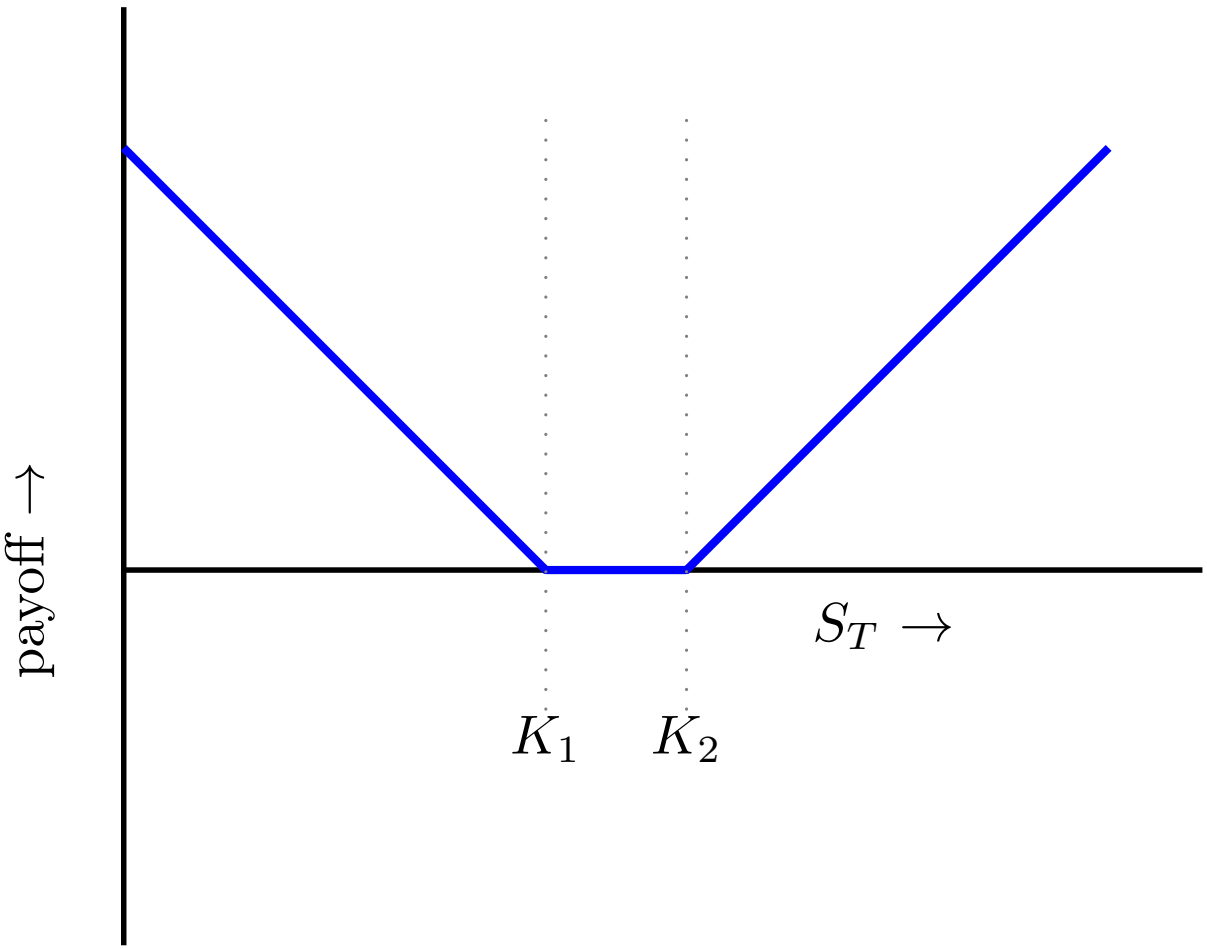
PAYOFF OF BUTTERFLY SPREAD



PAYOFF OF STRADDLE



PAYOFF OF STRANGLE



CORPORATE SECURITIES

CORPORATE SECURITIES SET UP

Suppose firm has risky assets worth V

Suppose it has debt (zero-coupon bond) with face value F

If $V \geq F$ then the firm repays its debt

If $V < F$, then the firm defaults; creditors seize the firm

Absent bankruptcy costs, the creditors get V if $V < F$

Thus the payoff of the creditors is

$$D_T = \min\{F, V\}$$

CORPORATE SECURITIES EXERCISE

Suppose that the assets V of a firm are traded

Assume one period and show by NA that for any F, if

$$D_T^F = \min\{F, V\}$$

and equity is the residual $E_T^F = V_T - D_T^F$ then for any F_1 and F_2

$$D_0^{F_1} + E_0^{F_1} = D_0^{F_2} + E_0^{F_2}.$$

Comment on the connection with MM Proposition 1

CONVERTIBLE BOND EXAMPLE

Convertible bonds allow the holder to convert into r shares

r is called the conversion ratio

CONVERTIBLE EXAMPLE

Suppose for simplicity that there is one bond only, CB

face value F and right to seize all assets in default

converts to $r\%$ of asset value if converted

Draw the payoff diagram

Write down the payoff

Replicate CB with V and liquid options on V

CONVERTIBLE SOLUTION

Payoff is the maximum of value of debt and conversion value

$$\begin{aligned} \text{CB}_T &= \max\left\{D_T, rV_T\right\} \\ &= \max\left\{\min\{V_T, F\}, rV\right\} \end{aligned}$$

Replicate this:

First, replicate debt

$$\begin{aligned} \min\{V_T, F\} &= F - (F - V)^+ \\ &= V - (V - F)^+ \end{aligned}$$

Recall that debt is "short an option" (on the assets)

What does this say about debt-holders' risk attitude?

CONVERTIBLE SOLUTION

To replicate the convertible, add r calls to this

What is the strike K? Convert if

$$rV > F$$

so the strike is F/r

Thus,

$$\mathrm{CB}_T = F - (F - V)^+ + r(V - F/r)^+$$

and

$$\mathrm{CB}_0 = FB_0 - p(F) + rc(F/r)$$

A convertible is a portfolio long F bonds, short a put, long r calls

RECAP

LECTURE 7: RECAP

Replicating securities

Portfolios of options

Corporate securities

FIN